

BEISEKER AGCM, AB, Canada

WMO: 710090

Lat: 51.379N

Lon: 113.357W

Elev: 896

StdP: 91.01

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.9	-27.3	-34.1	0.2	-30.9	-30.3	0.3	-27.1	12.4	-5.2	11.2	-7.5	1.8	350	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	29.2	16.6	27.2	15.8	25.4	15.3	18.9	25.5	17.7	24.3	16.6	23.1	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	13.1	22.6	15.2	12.0	21.0	13.9	11.1	19.4	57.5	25.4	53.5	24.3	49.8	23.1	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.8	8.4	-35.6	33.0	3.6	2.1	-38.2	34.5	-40.3	35.8	-42.3	36.9	-45.0	38.5		
(5)			-35.7	22.4	3.6	1.6	-38.3	23.5	-40.4	24.4	-42.4	25.3	-45.0	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-9.2	-10.1	-3.7	3.1	9.9	13.8	16.0	15.4	10.9	4.1	-3.6	-10.1		
	DBStd	11.35	8.28	8.18	7.79	5.00	4.21	2.94	2.48	2.97	4.52	4.88	6.92	8.05		
	HDD10.0	3121	594	563	426	212	54	4	0	2	42	191	409	624		
	HDD18.3	5579	853	797	684	458	263	139	79	98	226	442	658	882		
	CDD10.0	603	0	0	0	3	51	118	186	169	68	7	1	0		
	CDD18.3	19	0	0	0	0	1	3	7	7	2	0	0	0		
	CDH23.3	912	0	0	0	6	70	100	272	304	154	6	0	0		
	CDH26.7	222	0	0	0	1	8	17	66	88	42	1	0	0		
	WSAvg	3.3	3.4	3.2	3.5	3.9	4.1	3.8	2.9	2.7	3.1	3.2	3.1	2.9		
	PrecAvg	516	19	16	25	38	68	99	73	63	45	26	24	20		
	PrecMax	678	37	32	54	87	110	254	160	114	110	51	60	44		
(15) Precipitation	PrecMin	327	4	3	8	15	24	44	28	12	7	8	5	6		
	PrecStd	68	8	7	10	17	25	39	31	24	24	11	12	10		
	DB	7.9	7.6	15.9	23.1	27.7	28.7	30.9	32.1	30.5	23.4	16.1	8.6			
	MCWB	3.6	3.5	7.5	10.0	12.8	16.2	19.0	16.5	14.7	11.4	7.7	2.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	5.3	5.0	12.2	19.1	25.4	26.0	28.7	28.9	27.6	19.1	11.4	5.4			
	2% MCWB	2.2	1.8	5.2	8.7	12.2	15.1	18.1	16.8	14.0	9.9	5.5	2.0			
	5% DB	4.1	3.0	9.2	15.7	22.8	23.9	26.3	26.7	24.7	16.4	8.2	3.3			
	5% MCWB	1.5	0.6	3.7	6.6	11.2	14.3	17.6	16.2	13.2	8.4	3.3	0.5			
	10% DB	2.3	1.0	6.0	12.7	20.2	22.1	24.3	24.4	21.5	13.5	5.5	1.1			
	10% MCWB	0.2	-0.8	2.3	5.3	10.2	13.3	16.7	15.7	12.2	7.0	2.0	-1.0			
	0.4% WB	3.6	4.0	8.1	10.9	13.6	18.6	21.6	19.6	16.0	12.3	8.3	4.0			
	0.4% MCDB	7.1	7.3	14.9	21.7	24.9	25.2	27.4	25.8	26.9	21.6	15.5	7.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	2.5	2.1	5.9	8.8	12.6	16.7	19.6	18.3	14.8	10.4	5.7	2.3			
	2% MCDB	5.3	4.7	11.1	18.4	23.1	23.7	25.9	25.4	24.8	18.1	10.8	5.1			
	5% WB	1.6	0.7	3.9	7.1	11.7	15.3	18.3	17.3	13.6	8.8	3.9	0.5			
	5% MCDB	4.0	2.7	8.6	14.4	21.2	21.6	24.6	24.2	23.1	15.4	7.7	3.0			
	10% WB	0.2	-0.6	2.4	5.7	10.7	14.1	17.2	16.1	12.5	7.3	2.2	-1.1			
	10% MCDB	2.3	1.0	5.7	11.7	18.6	20.4	23.2	23.1	21.1	12.9	5.0	1.2			
	MDBR	11.1	11.4	11.8	13.6	14.7	13.6	15.2	16.0	16.3	14.8	12.1	11.1			
	5% DB	11.3	12.8	16.0	20.2	20.3	17.8	19.4	21.2	23.0	19.6	15.6	12.4			
(36) Mean Daily Temperature Range	MCWBR	8.5	9.9	9.5	10.7	8.9	8.6	10.9	10.2	10.6	10.9	9.8	9.3			
	MCDBR	11.0	12.0	15.2	18.8	17.6	15.6	17.8	18.2	21.1	18.6	15.0	12.2			
	MCWBR	8.4	9.5	9.6	10.1	7.9	8.4	10.6	9.9	10.0	10.5	9.9	9.4			
	MCDBR	11.1	12.8	16.0	20.2	20.3	17.8	19.4	21.2	23.0	19.6	15.6	12.4			
(40) Clear-Sky Solar Irradiance	taub	0.258	0.270	0.281	0.332	0.349	0.346	0.346	0.362	0.317	0.281	0.252	0.238			
	taud	2.374	2.423	2.471	2.342	2.352	2.401	2.411	2.365	2.451	2.534	2.487	2.383			
	Ebn at Noon	782	876	931	907	904	908	902	867	875	855	795	753			
	Edh at Noon	59	75	87	112	116	112	109	109	88	67	52	49			
(44) All-Sky Solar Radiation	RadAvg	1.11	2.13	3.50	4.75	5.74	6.17	6.60	5.33	3.76	2.27	1.22	0.87			
	RadStd	0.08	0.18	0.29	0.37	0.38	0.37	0.33	0.38	0.39	0.27	0.10	0.09			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page